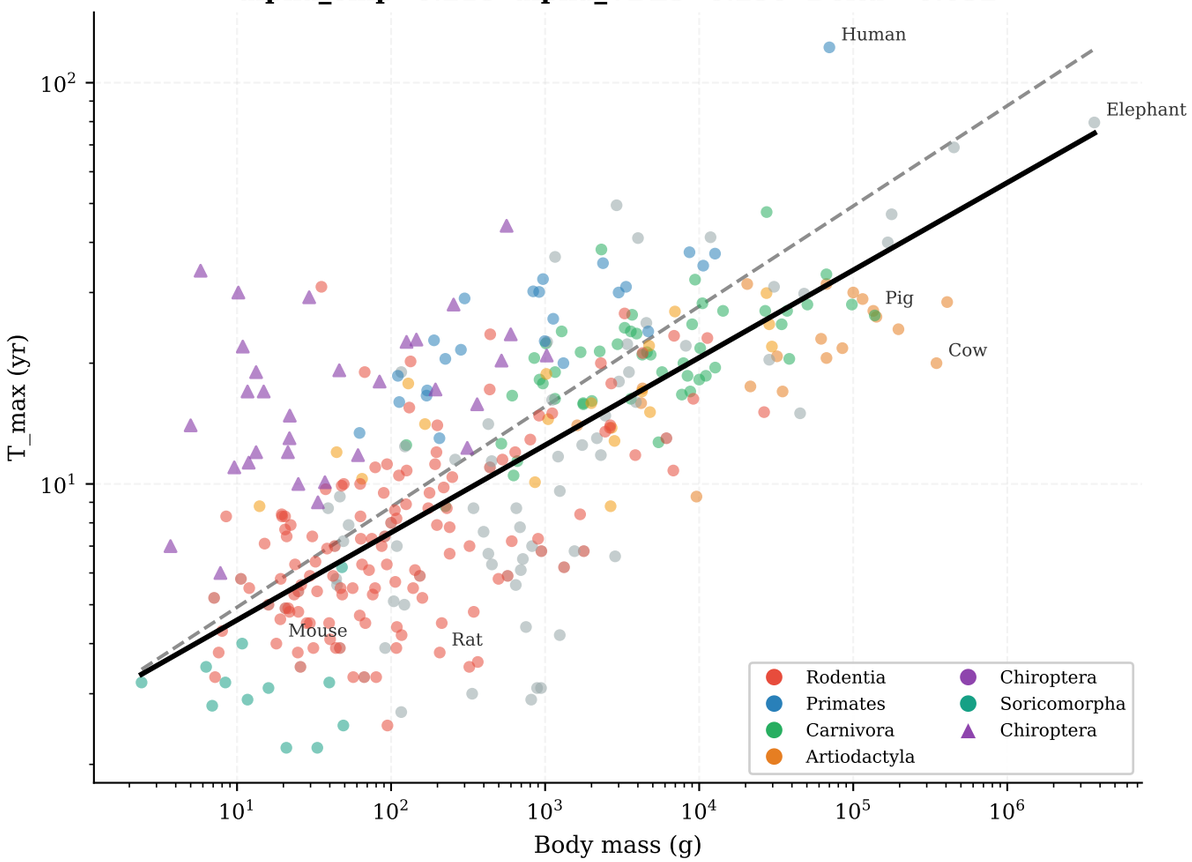
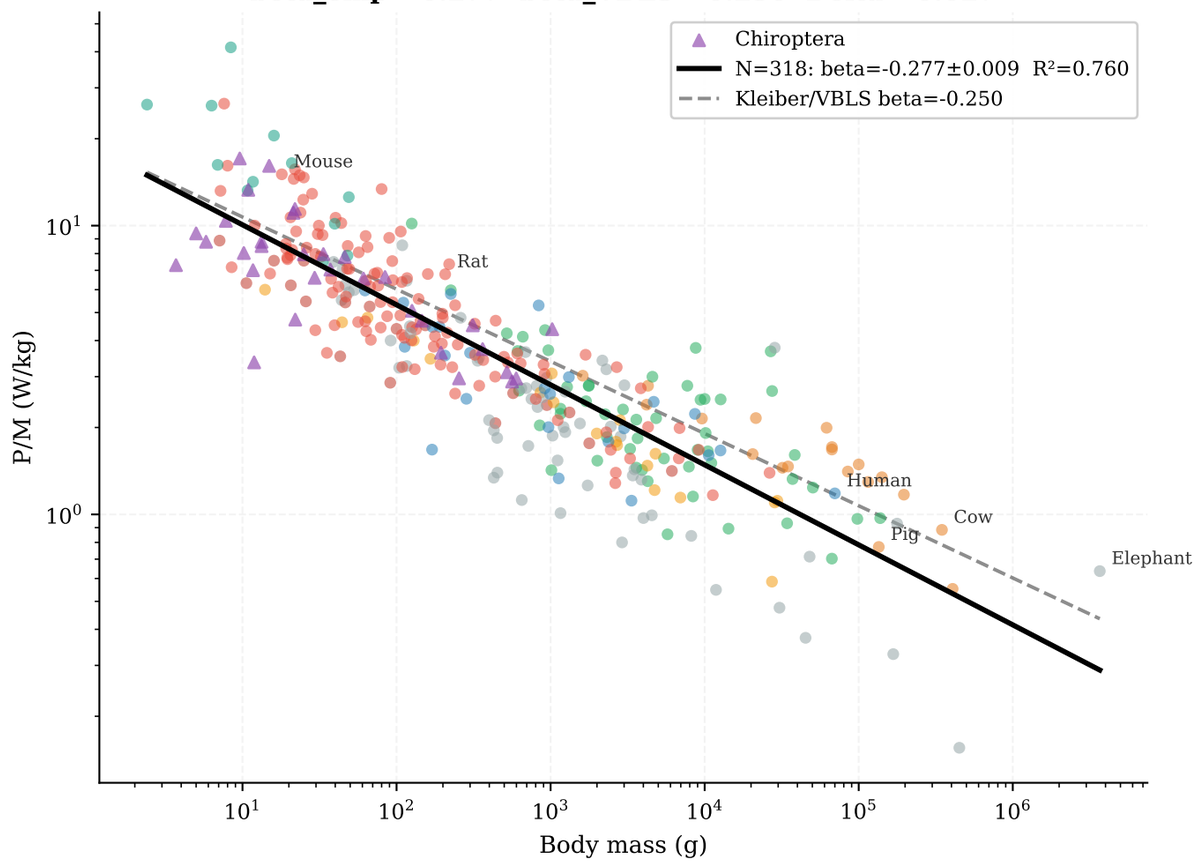
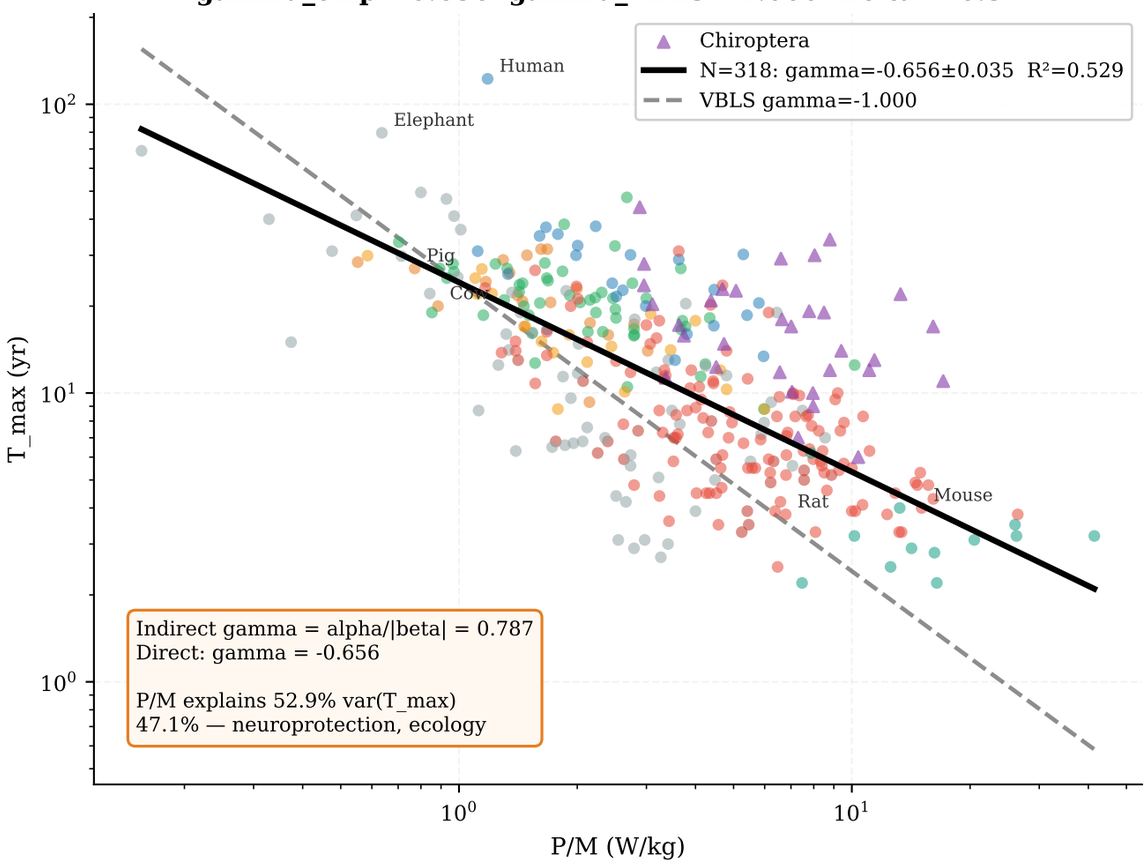
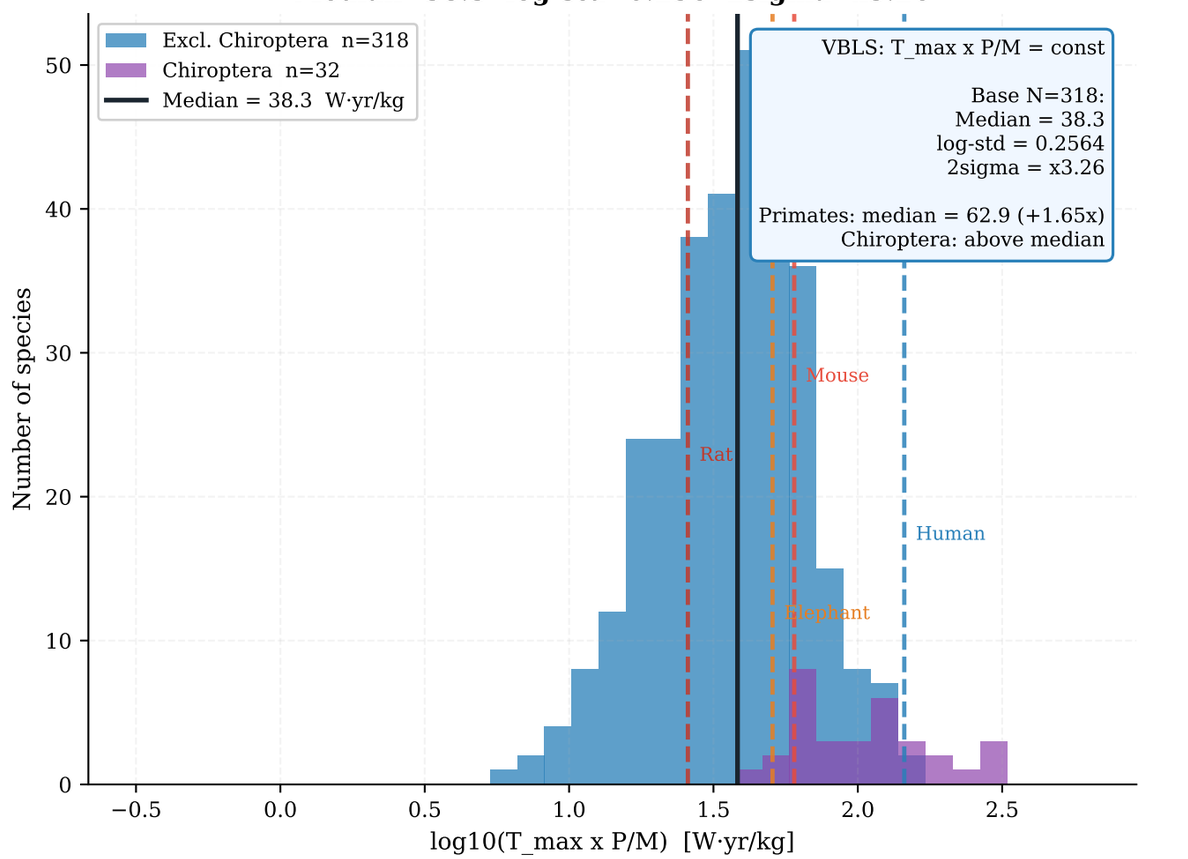

A. T_max ~ M^alpha
alpha_emp=0.218 alpha_VBLS=0.250 Delta=-0.032
T_max (yr)
Body mass (g)
Human
Elephant
Pig
Cow
Mouse
Rat
Rodentia
Primates
Carnivora
Artiodactyla
Chiroptera
Soricomorpha
Chiroptera


B. P/M ~ M^beta (Kleiber law)
beta_emp=-0.277 beta_VBLS=-0.250 Delta=-0.027
Chiroptera
N=318: beta=-0.277±0.009 R²=0.760
Kleiber/VBLS beta=-0.250
P/M (W/kg)
Body mass (g)
Mouse
Rat
Human
Cow
Pig
Elephant


C. T_max ~ (P/M)^gamma — key VBLS test
gamma_emp=-0.656 gamma_VBLS=-1.000 Delta=+0.344
Chiroptera
N=318: gamma=-0.656±0.035 R²=0.529
VBLS gamma=-1.000
Indirect gamma = alpha/|beta| = 0.787
Direct: gamma = -0.656
P/M explains 52.9% var(T_max)
47.1% — neuroprotection, ecology
T_max (yr)
P/M (W/kg)
Human
Elephant
Pig
Cow
Rat
Mouse


D. Rubner invariant: T_max x P/M = const
Median=38.3 log-std=0.256 2sigma=x3.26
Excl. Chiroptera n=318
Chiroptera n=32
Median = 38.3 W·yr/kg
VBLS: T_max x P/M = const
Base N=318:
Median = 38.3
log-std = 0.2564
2sigma = x3.26
Primates: median = 62.9 (+1.65x)
Chiroptera: above median
Number of species
log10(T_max x P/M) [W·yr/kg]
Mouse
Rat
Human
Elephant